

Siyakha Muli

Exercise 01

SQL Aggregate functions and Operations

Q1 Distinct departments

department
IT
HR
Finance

```
SELECT DISTINCT department  
FROM Students;
```

Q2 Average age per department

```
SELECT department, AVG(age) AS avg-age  
FROM Students  
GROUP BY department;
```

department	avg-age
IT	20.5
HR	22
Finance	23

Q3 Departments with more than 1 student

```
SELECT department, COUNT(*) AS student_count  
FROM Students  
GROUP BY department  
HAVING COUNT(*) > 1;
```

department	Student_Count
IT	2
HR	2

Q4 - Students aged between 21 and 23

```

SELECT student_id, name, age, department
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5 - IT or HR students older than 21

```

SELECT student_id, name, age, department
FROM Students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6 - Total credits per dept, only > 5 total (courses)

```
SELECT department, sum(credits) AS total_credits
FROM Courses
GROUP BY department
HAVING sum(credits) > 5;
```

department	total_credits
IT	11

Q7 - Courses without 4 credits

```
SELECT course_id, course_name, department, credits
FROM Courses
WHERE credits <> 4;
```

Course_id	Course_name	department	credits
101	SQL basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8 - Top 3 Courses by credits descending

```
SELECT course_id, course_name, credits
FROM Courses
ORDER BY credits DESC
LIMIT 3;
```

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

Q9, Max, min, avg. grade enrollments

```

SELECT MAX (grade) AS max_grade
       MIN (grade) AS min_grade
       AVG (grade) AS avg_grade
FROM enrollments;

```

Max - grade	min - grade	avg - grade
90	78	84.6

Q10 - Enrollment Count per Course

```

SELECT Course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY Course_id;

```

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Q11 - Total Salary and bonus per department (Salaries)

```
SELECT department,  
       sum (Salary) AS total - Salary  
       sum (Bonus) AS total - bonus  
FROM Salaries  
GROUP BY department;
```

department	total - Salary	total - bonus
IT	121 000	10500
HR	104 000	7500
Finance	70 000	6000

Q12 - Departments with avg. salary above 55 000

```
SELECT department, AVG (Salary) AS avg - salary  
FROM Salaries  
GROUP BY department  
HAVING AVG (Salary) > 55 000 ;
```

department	avg - salary
IT	61 000
Finance	70 000

Q13 - Employees where salary + bonus > 60 000

```
SELECT employee_id, name, salary, bonus,  
       'salary + bonus' AS  
       total - compensation  
FROM Salaries  
WHERE salary + bonus > 60 000;
```

employee - id	name	salary	bonus	Total - Compensation
1	Tom	60 000	8 000	68 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

Q 14 - Total and avg - budget, avg > 70 000 (projects)

```

SELECT department,
       sum (budget) AS total - budget
       AVG (budget) AS avg - budget
FROM   projects
GROUP BY department
HAVING AVG (budget) > 70 000;

```

Department	total - budget	avg - budget
IT	270 000	13 5000
Finance	80 000	80 000

Q 15 - Budget 50 000 - 120 000, not Marketing

```

SELECT project - id, project - name, department, budget
FROM   projects
GROUP BY department
WHERE  budget BETWEEN 50 000 AND 120 000
AND    department LIKE 'Marketing';

```

Project - id	Project - name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	60 000
5	HR Portal	HR	60 000